Controller performance comparison — numerical results

Numerical P, PI, PD and PID comparison for the specified spring–mass–damper plant (1 kg, 10 N·s/m, 20 N/m), 1 m target and 0.001 s time step.

Controller	Rise time (s)	Overshoot (%)	Settling time (s)	Final error (m)
P	0.142	0.000	Not settled	0.167
PI	0.172	4.510	5.820	0.005
PD	0.605	0.000	Not settled	0.167
PID	0.875	0.000	3.511	0.002

Metric definitions. Rise time is measured from 10% to 90% of the achieved final output value. Overshoot is the peak amount above the 1 m target as a percentage of target. Settling time is the last time the response enters and remains within ±2% of the target. Final error is target minus the last simulated position.

Not settled. A response is marked “Not settled” if it does not remain within the ±2% target band by the end of the simulation. P and PD retain an expected steady-state offset because they lack integral action.

Gain set. P: Kp=100; PI: Kp=100, Ki=40; PD: Kp=100, Kd=25; PID: Kp=100, Ki=40, Kd=25. Values are illustrative for this model.

Reproduction. Run `python src/run_all.py` from the repository root. The source prints these metrics and regenerates all figures.